ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
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17.10CP Structural Checklist for Building Type CFS2: Cold-formed Steel Lightframe Bearing Wall Construction, Strap-Braced Lateral Wall System

Table 17-20. Collapse Prevention Structural Checklist for Building Type CFS2.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low and Moderate Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of strap-braced walls in each principal direction is greater than or equal to 2.	5.5.1.1	A.3.2.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAP-BRACED WALLS—BRACE AXIAL STRESS CHECK: The axial stress in the diagonal straps, calculated using the Quick Check procedure of Section 4.4.3.4 is less than $0.50F_y$.	5.5.4.1	A.3.3.1.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAP-BRACED WALLS—CHORD STUD AXIAL CHECK: The axial force caused by overturning plus the gravity load on the end stud is less than the nominal strength of the end stud calculated in accordance with AISI S100.	5.5.4.9.5	A.3.3.2.9	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	NARROW STRAP-BRACED WALLS: Narrow strap-braced walls with an aspect ratio greater than 2-to-1 are not used to resist seismic forces.	5.5.4.9.1	A.3.3.2.5	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	WALLS CONNECTED THROUGH FLOORS: Strap-braced walls have an interconnection between stories to transfer overturning and shear forces through the floor.	5.5.4.9.2	A.3.3.2.6	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	HILLSIDE SITE: For structures that are taller on at least one side by more than one-half story because of a sloping site, all strap-braced walls on the downhill slope have an aspect ratio less than 1-to-1.	5.5.4.9.3	A.3.3.2.7	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	POSTS: There is a positive connection of posts to the foundation.	5.7.3.3	A.5.3.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SILLS (BASE TRACK): All sills or base tracks are bolted to the foundation.	5.7.3.3	A.5.3.4	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	GIRDER–COLUMN CONNECTION: There is a positive connection using plates, connection hardware, or straps between the girder and the column support.	5.7.4.1	A.5.4.1	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Low and Moderate Seismicity)							
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SILL (BASE TRACK) BOLTS: Bolts are spaced at 6 ft (1.8 m) or less with acceptable edge and end distance provided for steel and concrete.	5.7.3.3	A.5.3.7	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STRAP-BRACE CONNECTIONS: Strap connections develop the yield capacity of the straps.	5.5.4.4	A.3.3.1.5	
Diaphragms							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DIAPHRAGM CONTINUITY: Floor and roof diaphragms do not have expansion joints or vertical offsets, such as split levels, sawtooth, or clerestory configurations.	5.6.1.1	A.4.1.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	ROOF CHORD CONTINUITY: All chord elements are continuous, regardless of changes in roof elevation.	5.6.1.1	A.4.1.3	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	SPANS: All diaphragms with spans greater than 24 ft (7.3 m) consist of wood structural panels.	5.6.2	A.4.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	UNBLOCKED DIAPHRAGMS: All unblocked wood structural panel diaphragms have horizontal spans less than 40 ft (12.2 m) and have aspect ratios less than or equal to 4-to-1.	5.6.2	A.4.2.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1	

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